

4. Iterated Frobenius morphism; perfect preschemes

For every integer $\nu \geq 0$, the iterated endomorphism fr_X^ν depends functorially on the prescheme X of characteristic p . Thus, if S is a prescheme of characteristic p and X is an S -prescheme, fr_X^ν factors, in a unique way, as an S -morphism

$$\mathrm{Fr}_{X/S}^\nu : X \longrightarrow X^{(p^\nu/S)},$$

where

$$X^{(p^\nu/S)} = X \times_S (S, \mathrm{fr}_S^\nu),$$

followed by the projection $X^{(p^\nu/S)} \rightarrow X$. The morphism $\mathrm{Fr}_{X/S}^\nu$ depends functorially on the S -prescheme X and is compatible with change of the base S (cf. no. 2, prop. 1b)). Moreover it is clear that

$$X^{(p^\nu/S)} = X^{(p^\nu)} \simeq (X^{(p^{\nu-1})})^{(p)}$$

and that

$$\mathrm{Fr}_{X/S}^\nu = (\mathrm{Fr}_{X/S}^{\nu-1})^{(p)} \circ \mathrm{Fr}_{X/S},$$

modulo the identification allowed by the preceding canonical isomorphism.

Definition 4. A prescheme S of characteristic p is said to be perfect if fr_S is an automorphism of S .

If S is a perfect prescheme of characteristic p and X is an S -prescheme, one sees immediately that $X^{(p/S)}$ is identified with the prescheme X considered as an S -prescheme by means of the morphism $\mathrm{fr}_S^{-1} \circ g : X \rightarrow S$, where g is the structural morphism of X into S ; modulo this identification, $\mathrm{Fr}_{X/S} = \mathrm{fr}_X$.

As an example of a perfect prescheme, we shall meet $S = \mathrm{Spec}(\mathbb{F}_q)$, where $q = p^\nu$ is a power of the characteristic p ; indeed, in this case $\mathrm{fr}_S^\nu = \mathrm{id}_S$. If X is an S -prescheme, one sees that

$$X^{(p^\nu/S)} = X, \quad \mathrm{Fr}_{X/S}^\nu = \mathrm{fr}_X^\nu.$$

Note that for $\nu = 1$ one obtains $S = \mathrm{Spec}(\mathbb{F}_p)$, for which $\mathrm{fr}_S = \mathrm{id}_S$; the S -preschemes are the preschemes of characteristic p ; over this base S , $X^{(p/S)} = X$ and $\mathrm{Fr}_{X/S} = \mathrm{fr}_X$.

§ 2. Frobenius correspondence

1. Definition

Let X be a prescheme of characteristic p , and let F be a sheaf of sets on $X_{\mathrm{\acute{e}t}}$. For every étale X -prescheme U one has

$$(\mathrm{fr}_X)_*(F)(U) = F(\mathrm{fr}_X^{-1}(U)) = F(U^{(p/X)}),$$

and the Frobenius morphism $\mathrm{Fr}_{U/X} : U \rightarrow U^{(p/X)}$ defines a map

$$F(\mathrm{Fr}_{U/X}) : \mathrm{fr}_{X*}(F)(U) \longrightarrow F(U),$$

which is visibly functorial in U (since $\mathrm{Fr}_{U/X}$ is functorial in U). Thus one obtains a morphism of sheaves on $X_{\text{ét}}$:

$$(\mathrm{fr}_X)_*(F) \longrightarrow F.$$

Now, when U is étale over X , $\mathrm{Fr}_{U/X}$ is an isomorphism (§ 1 no. 2, prop. 2c)), hence the same is true of $F(\mathrm{Fr}_{U/X})$; it follows that the preceding morphism is an isomorphism. In the case where F is endowed with an algebraic structure, for example if it is a sheaf of Λ -modules, where Λ is a given base ring, this isomorphism is compatible with the algebraic structure. The inverse

$$F(\mathrm{Fr}_{/X})^{-1} : F \longrightarrow \mathrm{fr}_{X*}(F)$$

of this isomorphism defines, since $(\mathrm{fr}_X)^*$ is left adjoint to the functor $(\mathrm{fr}_X)_*$, a morphism

$$\mathrm{Fr}_{F/X}^* : (\mathrm{fr}_X)^*(F) \longrightarrow F.$$

One can even say more: since fr_X is a universal homeomorphism (§ 1 no. 2, prop. 2a)), the functors $(\mathrm{fr}_X)^*$ and $(\mathrm{fr}_X)_*$ are quasi-inverse to each other; consequently $\mathrm{Fr}_{F/X}^*$ is an isomorphism, just as is the morphism $F(\mathrm{Fr}_{/X})^{-1}$ from which it comes.

Definition 1. Given a prescheme X of characteristic p and a sheaf of sets F on $X_{\text{ét}}$, the Frobenius correspondence on (X, F) is the pair $(\mathrm{fr}_X, \mathrm{Fr}_{F/X}^*)$ formed by the Frobenius endomorphism fr_X of X and the isomorphism

$$\mathrm{Fr}_{F/X}^* = (F(\mathrm{Fr}_{/X})^{-1})^\# : (\mathrm{fr}_X)^*(F) \longrightarrow F.$$

Of course one defines in the same way a class of iterated Frobenius correspondence $(\mathrm{fr}_X^\nu, (\mathrm{Fr}_{F/X}^*)^\nu)$ for every integer $\nu \geq 0$; the isomorphism

$$(\mathrm{Fr}_{F/X}^*)^\nu : (\mathrm{fr}_X^\nu)^*(F) \longrightarrow F$$

is equal to

$$(\mathrm{Fr}_{F/X}^*)^{\nu-1} \circ (\mathrm{fr}_X^{\nu-1})^*(\mathrm{Fr}_{F/X}^*)$$

or also to

$$\mathrm{Fr}_{F/X}^* \circ \mathrm{fr}_X^*((\mathrm{Fr}_{F/X}^*)^{\nu-1})$$

for $\nu \geq 1$.

Let us examine the particular case where F is representable by an étale prescheme V over X ; then $(\mathrm{fr}_X)^*(F) = V^{(p/X)}$, $F(V) = \mathrm{Hom}_X(V, V)$,

$$(\mathrm{fr}_X)^*(F)(V) = F(V^{(p/X)}) = \mathrm{Hom}_X(V^{(p/X)}, V),$$

and $F(\mathrm{Fr}_{V/X})$ is the map from $\mathrm{Hom}_X(V^{(p/X)}, V)$ to $\mathrm{Hom}_X(V, V)$ which sends a morphism ψ to $\psi \circ \mathrm{Fr}_{V/X}$. Thus $\mathrm{Fr}_{F/X}^*$ is identified with a morphism from $V^{(p/X)}$ to V , which is the image of id_V under the inverse map $F(\mathrm{Fr}_{V/X})^{-1}$ of the preceding one. This shows that for $F = V$ étale over X one has

$$\mathrm{Fr}_{V/X}^* = (\mathrm{Fr}_{V/X})^{-1}.$$

Note that if the sheaf F is constant, then it is clear that

$$(\mathrm{fr}_X)^*(F) = F, \quad \mathrm{Fr}_{F/X}^* = \mathrm{id}_F.$$

Proposition 1.

- a) The isomorphism $\mathrm{Fr}_{F/X}^*$ depends functorially on the sheaf F ; in other words there is a functorial isomorphism

$$\mathrm{Fr}_{F/X}^* \mathrm{fr}_X^* \longrightarrow \mathrm{id}_{\widehat{X}_{\mathrm{\acute{e}t}}},$$

where $\widehat{X}_{\mathrm{\acute{e}t}}$ is the category of sheaves of sets on $X_{\mathrm{\acute{e}t}}$.

- b) The Frobenius correspondence $(\mathrm{fr}_X, \mathrm{Fr}_{F/X}^*)$ on (X, F) is compatible with change of the base X . That is, if $\mathcal{F}$ denotes the fibred category over $(\mathrm{Sch})_p$ of sheaves, for the étale topology, on variable preschemes of characteristic p , the functors fr_X^* , $X \in \mathrm{ob}(\mathrm{Sch})_p$, define a cartesian $(\mathrm{Sch})_p$ -functor

$$\mathrm{fr}^* : \mathcal{F} \longrightarrow \mathcal{F},$$

and the morphisms $\mathrm{Fr}_{F/X}^*$, $X \in \mathrm{ob}(\mathrm{Sch})_p$, $F \in \mathrm{ob} \widehat{X}_{\mathrm{\acute{e}t}}$, define a functorial $(\mathrm{Sch})_p$ -morphism

$$\mathrm{Fr}^* : \mathrm{fr}^* \longrightarrow \mathrm{id}_{\mathcal{F}}$$

which is an isomorphism.

- c) Conversely, these properties characterize Fr^* ; that is, there exists a unique functorial $(\mathrm{Sch})_p$ -morphism from fr^* to $\mathrm{id}_{\mathcal{F}}$.

Notice that statement b) implies statement a); the latter is entirely evident.

Let us prove b). Let $g : X' \rightarrow X$ be a morphism of preschemes of characteristic p , and let F be a sheaf on $X_{\mathrm{\acute{e}t}}$. Put $F' = g^*(F)$ and, more generally, mark by a prime the inverse image by g of any object given over X . Since

$$g \circ \mathrm{fr}_{X'} = \mathrm{fr}_X \circ g,$$

one has

$$\mathrm{fr}_{X'}^*(F') = \mathrm{fr}_{X'}^*(g^*F) = g^*(\mathrm{fr}_X^*F) = (\mathrm{fr}_X^*F)',$$

whence the fact that the fr_X^* define a cartesian functor fr^* over $(\mathrm{Sch})_p$. It remains to prove that

$$\mathrm{Fr}_{F'/X'}^* = (\mathrm{Fr}_{F/X}^*)'.$$

Using the properties of adjoint functors, one sees that this amounts to showing that the image of $F(\mathrm{Fr}_{X'}^*)^{-1}$ under the bijection

$$\mathrm{Hom}_{X'}(F', \mathrm{fr}_{X'*}(F')) = \mathrm{Hom}_{X'}(g^*F, \mathrm{fr}_{X'*}g^*F) \simeq \mathrm{Hom}_X(F, g_*\mathrm{fr}_{X'*}g^*F)$$

coincides with the image of $F(\mathrm{Fr}_X^*)^{-1}$ under the map

$$\mathrm{Hom}_X(F, \mathrm{fr}_{X*}F) \longrightarrow \mathrm{Hom}_X(F, g_*\mathrm{fr}_{X'*}g^*F)$$

defined by the adjunction morphism $F \rightarrow g_*g^*F$; in other words, one must show that

$$\alpha \circ F(\mathrm{Fr}_X^*) = g_*(F'(\mathrm{Fr}_{X'}^*)) \circ \mathrm{fr}_{X*}(\alpha),$$

where $\alpha : F \rightarrow g_*g^*F$ is the adjunction. This says that for every prescheme U étale over X there is a commutative diagram

$$\begin{array}{ccc} F(U(p/X)) & \xrightarrow{F(\mathrm{Fr}_{U/X})} & F(U) \\ \downarrow & & \downarrow \\ F'(U'(p/X')) & \xrightarrow{F'(\mathrm{Fr}_{U'/X'})} & F'(U'), \end{array}$$

where the vertical arrows are the canonical maps; but this follows immediately from

$$\mathrm{Fr}_{U'/X'} = (\mathrm{Fr}_{U/X})'$$

(§ 1, no. 2, prop. 1b)).

To prove c), after composing with $(\mathrm{Fr}^*)^{-1}$, we are reduced to proving that $\mathrm{id}_{\mathcal{F}}$ has only one $(\mathrm{Sch})_p$ -endomorphism, namely the identity. If ψ is a $(\mathrm{Sch})_p$ -endomorphism of $\mathrm{id}_{\mathcal{F}}$, let F be a sheaf on a prescheme X of characteristic p , and let U be an étale prescheme over X . We know that $F(U)$ is identified with $\mathrm{Hom}_U(U, F|U)$, and $\psi_F(U) : F(U) \rightarrow F(U)$ with the map

$$s \mapsto \psi_{F|U} \circ s = s \circ \psi_U,$$

where $s \in \mathrm{Hom}_U(U, F|U)$; since the final sheaf U of $\tilde{U}_{\mathrm{ét}}$ has no endomorphism other than the identity, one has $\psi_U = \mathrm{id}_U$. It follows that $\psi_F(U)$ is also the identity of $F(U)$, and hence that $\psi = \mathrm{id}_{\mathcal{F}}$.

Of course, if the sheaf F is endowed with an algebraic structure, the isomorphism $\mathrm{Fr}_{F/X}^*$ is compatible with this structure; for example, if F is a sheaf of groups on the prescheme X of characteristic p , $\mathrm{Fr}_{F/X}^*$ is an isomorphism of sheaves of groups. Let Λ be a ring and let F be a sheaf of Λ -modules on X ; then $\mathrm{Fr}_{F/X}^*$ is an isomorphism of sheaves of Λ -modules. More generally, if $F^\bullet$ is a complex of Λ -modules, one immediately defines an isomorphism

$$\mathrm{Fr}_{F^\bullet/X}^* : \mathrm{fr}_X^*(F^\bullet) \longrightarrow F^\bullet$$

which is equal to $\mathrm{Fr}_{F^i/X}^*$ on the component of degree i , thanks to proposition 1a). In this case the pair $(\mathrm{fr}_X, \mathrm{Fr}_{F^\bullet/X}^*)$ is indeed a correspondence, in the usual sense, on $(X, F^\bullet)$; this correspondence is functorial in $F^\bullet$ and compatible with change of the base X .

2. Effect on cohomology

Proposition 2. Let X be a prescheme of characteristic p .

- a) For every sheaf of sets F on $X_{\mathrm{ét}}$, the endomorphism of $H^0(X, F)$ defined by $(\mathrm{fr}_X, \mathrm{Fr}_{F/X}^*)$, that is, the composite

$$H^0(X, F) \longrightarrow H^0(X, \mathrm{fr}_X^* F) \longrightarrow H^0(X, F),$$

where the first arrow is the canonical map arising from the contravariant character of $H^0(X, F)$ with respect to X , and the second is $H^0(X, \mathrm{Fr}_{F/X}^*)$, is the identity of $H^0(X, F)$.

- b) For every sheaf of groups F on $X_{\mathrm{ét}}$, the endomorphism of $H^1(X, F)$ defined by $(\mathrm{fr}_X, \mathrm{Fr}_{F/X}^*)$, that is,

$$H^1(X, F) \longrightarrow H^1(X, \mathrm{fr}_X^* F) \longrightarrow H^1(X, F),$$

is the identity.

- c) For every left bounded complex F of Λ -modules on $X_{\mathrm{ét}}$, the endomorphism of $\mathrm{R}\Gamma(X, F)$ defined by $(\mathrm{fr}_X, \mathrm{Fr}_{F/X}^*)$, that is, the composite

$$\mathrm{R}\Gamma(X, F) \longrightarrow \mathrm{R}\Gamma(X, \mathrm{fr}_X^* F) \longrightarrow \mathrm{R}\Gamma(X, F),$$

is the identity.

We first establish a); b) and c) will be deduced from it. Let ψ_F be the considered endomorphism of $H^0(X, F)$. It is clear that ψ_F depends functorially on F and that $\psi_X = \text{id}_{H^0(X, X)}$. Every element s of $H^0(X, F)$ is identified with a morphism of sheaves $s : X \rightarrow F$ such that s is the image under $H^0(X, s)$ of the unique element of $H^0(X, X)$; since

$$\psi_F \circ H^0(X, s) = H^0(X, s) \circ \psi_X = H^0(X, s),$$

by the preceding remarks, one sees that $\psi_F(s) = s$ for every s , hence $\psi_F = \text{id}_{H^0(X, F)}$.

Note that ψ_F is also the composite

$$H^0(X, F) \longrightarrow H^0(X, \text{fr}_{X*} F) \longrightarrow H^0(X, F),$$

where the first arrow is $H^0(X, F(\text{Fr}_X)^{-1})$ and the second is the canonical map, which is nothing but the identity. Similarly, the endomorphisms considered in b) and c) factor as

$$H^1(X, F) \longrightarrow H^1(X, \text{fr}_{X*} F) \longrightarrow H^1(X, F),$$

and

$$\text{R}\Gamma(X, F) \longrightarrow \text{R}\Gamma(X, \text{fr}_{X*} F) \longrightarrow \text{R}\Gamma(X, F).$$

To establish b), we shall use a monomorphism $F \rightarrow G$ of F into a sheaf of groups G such that $H^1(X, G) = 0$. Then $\text{fr}_{X*} F \rightarrow \text{fr}_{X*} G$ is a monomorphism and $H^1(X, \text{fr}_{X*} G) = 0$. Introduce the quotient homogeneous sheaf of sets $C = G/F$. Since fr_X is integral, surjective, and radicial, the functors fr_X^* and fr_{X*} are quasi-inverse equivalences of categories (SGA VIII theorem 1.1). It follows that

$$\text{fr}_{X*}(C) \simeq \text{fr}_{X*}(G)/\text{fr}_{X*}(F),$$

and the exact cohomology sequence (SGA XII proposition 3.1) shows that the considered endomorphism of $H^1(X, F)$ arises, by passage to the quotient, from

$$\psi_C : H^0(X, C) \longrightarrow H^0(X, C),$$

which is the identity by a). This endomorphism is therefore $\text{id}_{H^1(X, F)}$.

Assertion c) is proved analogously, using a complex I that is injective in each degree and isomorphic to F in the derived category D^+ , built from the Λ -modules. The complex $\text{fr}_{X*}(I)$ is injective in each degree and isomorphic to $\text{fr}_{X*}(F)$ in D^+ , since fr_{X*} is an equivalence of categories. One may then identify $\text{R}\Gamma(X, F)$ with $H^0(X, I)$ and $\text{R}\Gamma(X, \text{fr}_{X*} F)$ with $H^0(X, \text{fr}_{X*} I)$; one sees that the considered endomorphism of $\text{R}\Gamma(X, F)$ is identified with ψ_I , which is the identity. This endomorphism is therefore $\text{id}_{\text{R}\Gamma(X, F)}$.

3. Behavior of products

Let X and X' be preschemes of characteristic p . Their product $X \times X'$ is identical with the fibred product $X \times_e X'$, where $e = \text{Spec}(\mathbb{F}_p)$. It is clear that

$$\text{fr}_{X \times X'} = \text{fr}_X \times \text{fr}_{X'} = (\text{fr}_X \times \text{id}_{X'}) \circ (\text{id}_X \times \text{fr}_{X'}) = (\text{id}_X \times \text{fr}_{X'}) \circ (\text{fr}_X \times \text{id}_{X'}).$$

It follows immediately that, for every X -prescheme Y and every X' -prescheme Y' , there is a canonical isomorphism

$$(Y \times Y')^{(p/X \times X')} \simeq Y^{(p/X)} \times Y'^{(p/X')};$$

modulo this isomorphism one can write

$$\mathrm{Fr}_{Y \times Y'/X \times X'} = \mathrm{Fr}_{Y/X} \times \mathrm{Fr}_{Y'/X'}.$$

Proposition 3. Let X and X' be preschemes of characteristic p , let Λ be a commutative ring, let F be a sheaf of Λ -modules on $X_{\text{ét}}$, and let F' be a sheaf of Λ -modules on $X'_{\text{ét}}$. One has

$$\begin{aligned} \mathrm{Fr}_{F \boxtimes_{\Lambda} F'/X \times X'}^* &= \mathrm{Fr}_{F/X}^* \otimes \mathrm{Fr}_{F'/X'}^* \\ &= (\mathrm{Fr}_{F/X}^* \otimes \mathrm{id}_{\mathrm{fr}_{X'}^* F'}) \circ (\mathrm{id}_{\mathrm{fr}_X^* F} \otimes \mathrm{Fr}_{F'/X'}^*) \\ &= (\mathrm{id}_F \otimes \mathrm{Fr}_{F'/X'}^*) \circ (\mathrm{Fr}_{F/X}^* \otimes \mathrm{id}_{F'}). \end{aligned}$$

To write these equalities, one has identified

$$\mathrm{fr}_{X \times X'}^*(F \boxtimes_{\Lambda} F')$$

with the tensor product $\mathrm{fr}_X^*(F) \otimes_{\Lambda} \mathrm{fr}_{X'}^*(F')$ by means of the canonical isomorphism.

The two sides of the first equality depend functorially on F and F' , and are compatible with change of the base X or the base X' . Moreover they are compatible with passage to the inductive limit in F or in F' , since the tensor product commutes with inductive limits and it is clear that

$$\mathrm{Fr}_{(\varinjlim F_i)/X}^* = \varinjlim \mathrm{Fr}_{F_i/X}^*;$$

this permits reduction to the case where $F = X$ and $F' = \Lambda_{X'}$, in which case it is evident.

Corollary. With the data of proposition 3, the endomorphisms of $\mathrm{R}\Gamma_{X \times X'}(F \boxtimes F')$ defined respectively by

$$(\mathrm{id}_X \times \mathrm{fr}_{X'}, \mathrm{id}_F \otimes \mathrm{Fr}_{F'/X'}^*)$$

and by

$$(\mathrm{fr}_X \times \mathrm{id}_{X'}, \mathrm{Fr}_{F/X}^* \otimes \mathrm{id}_{F'})$$

are inverse isomorphisms of each other.

Denoting these endomorphisms by ψ' and ψ , let us show for example that $\psi \circ \psi' = \mathrm{id}$. This follows from the commutative diagram

$$\begin{array}{ccccc} \mathrm{R}\Gamma_{X \times X'}(F \boxtimes F') & \xrightarrow{\mathrm{id}_X \times \mathrm{fr}_{X'}} & \mathrm{R}\Gamma_{X \times X'}(F \boxtimes \mathrm{fr}_{X'}^* F') & \xrightarrow{\mathrm{R}\Gamma(\mathrm{id}_F \otimes \mathrm{Fr}_{F'/X'}^*)} & \mathrm{R}\Gamma_{X \times X'}(F \boxtimes F') \\ & \searrow \mathrm{fr}_{X \times X'} & \downarrow \mathrm{fr}_X \times \mathrm{id}_{X'} & & \downarrow \mathrm{fr}_X \times \mathrm{id}_{X'} \\ & & \mathrm{R}\Gamma_{X \times X'}(\mathrm{fr}_X^* F \boxtimes \mathrm{fr}_{X'}^* F') & \xrightarrow{\mathrm{R}\Gamma(\mathrm{Fr}_{F/X}^* \otimes \mathrm{id}_{F'})} & \mathrm{R}\Gamma_{X \times X'}(F \boxtimes F'), \end{array}$$

which is commutative because

$$\mathrm{fr}_{X \times X'} = (\mathrm{fr}_X \times \mathrm{id}_{X'}) \circ (\mathrm{id}_X \times \mathrm{fr}_{X'})$$

for the upper triangle and because

$$\mathrm{Fr}_{F \boxtimes F'/X \times X'}^* = (\mathrm{Fr}_{F/X}^* \otimes \mathrm{id}_{F'}) \circ (\mathrm{id}_{\mathrm{fr}_X^* F} \otimes \mathrm{Fr}_{F'/X'}^*)$$

for the lower triangle, and in which the composite of the two oblique arrows is the identity by proposition 2c). With an analogous diagram exchanging the roles of (X, F) and (X', F') , one obtains $\psi' \circ \psi = \text{id}$.

We shall apply this result to the particular case where $X' = e = \text{Spec}(\overline{\mathbb{F}}_p)$ is the spectrum of an algebraic closure $\overline{\mathbb{F}}_p$ of $\mathbb{F}_p$ and where $F' = \Lambda_e$ is the constant sheaf with fibre Λ . In this case we put

$$\overline{X} = X \times_e e = X \otimes_{\mathbb{F}_p} \overline{\mathbb{F}}_p, \quad \overline{F} = F \otimes_{\Lambda} \Lambda_e,$$

the inverse image of F by the projection $\overline{X} \rightarrow X$. The Frobenius endomorphism fr_e is identified with the canonical generator

$$f : \lambda \mapsto \lambda^p \quad (\lambda \in \overline{\mathbb{F}}_p)$$

of the Galois group

$$\pi_e = \text{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p),$$

where f determines an isomorphism $\widehat{\mathbb{Z}} \simeq \pi_e$, and

$$\text{Fr}_{\Lambda_e/e}^* = \text{id}_{\Lambda_e}$$

since Λ_e is a constant sheaf. Put

$$\overline{\text{fr}}_X = \text{fr}_X \times \text{id}_e,$$

the inverse image of fr_X by $\overline{X} \rightarrow X$,

$$f_X = \text{id}_X \times \text{fr}_e,$$

the automorphism of $\overline{X}$ defined by f by transport of structure, and

$$\overline{\text{Fr}}_{F/X}^* = \text{Fr}_{F/X}^* \otimes \text{id}_{\Lambda_e},$$

the inverse image of $\text{Fr}_{F/X}^*$ by $\overline{X} \rightarrow X$. We may write

$$\text{fr}_{\overline{X}} = \overline{\text{fr}}_X \circ f = f \circ \overline{\text{fr}}_X, \quad \text{Fr}_{\overline{F}/\overline{X}}^* = \overline{\text{Fr}}_{F/X}^*.$$

The corollary of proposition 3 shows that the endomorphism of $\text{R}\Gamma_{\overline{X}}(\overline{F})$ defined by the pair $(\text{fr}_X, \overline{\text{Fr}}_{F/X}^*)$ is the inverse automorphism of the one defined by f_X . We shall say that $(\text{fr}_X, \overline{\text{Fr}}_{F/X}^*)$ is the geometric Frobenius correspondence; it is the inverse image by $\overline{X} \rightarrow X$ of the Frobenius correspondence on (X, F) . We denote by

$$\text{fr}_{\text{R}\Gamma_{\overline{X}}(\overline{F})}$$

the automorphism of $\text{R}\Gamma_{\overline{X}}(\overline{F})$ which it defines. The operation of arithmetic Frobenius, coming by transport of structure from $f \in \pi_e$, is its inverse.

§ 3. The L -function

1. Definition

Let p be a prime number; consider a prime number $\ell \neq p$ and a finite extension Ω of $\mathbb{Q}_\ell$. To every pair (X, F) consisting of a scheme X of finite type over $e = \text{Spec}(\mathbb{F}_p)$ and a constructible Ω -sheaf F on X , we shall associate a formal series

$$L_F \in \Omega[[t]],$$

with constant term 1.